5 hours on examples

<u>Problem 1 (11JMO1)</u>

We claim that $n = 1$ is the only answer. We have $2 + 12 + 2011 = 2025 = 45^2$.

Clearly, $n = 0$ fails. FTSOC, let there be greater n.

Claim: n is even.

Taking mod 4 for $n \geq 2$,

$$2^n + 12^n + 2011^n \equiv 2011^n \equiv 3^n \bmod 4$$

Clearly, $n \equiv 0 \bmod 2$ since otherwise $3^n \equiv 3 \bmod 4$ is not a quadratic residue.

Claim: n is odd.

Taking mod 3,

$$2^n + 12^n + 2011^n \equiv 2^n + 1^n \bmod 3$$

Clearly, $n \equiv 1 \bmod 2$ since otherwise $2^n + 1^n \equiv 2 \bmod 3$ is not a quadratic residue.

The two contradict and we are done.

<u>Problem 3 (87AMO1)</u>

We have the solutions, $(m, n) = (-1, -1), (8, -10), (9, -6), (9, -21)$ or $n = 0$. These all clearly work.

Expanding,

$$m^3 + m^2n^2 + mn + n^3 = m^3 - 3m^2n + 3mn^2 - n^3$$

$$\Leftrightarrow m^2(n^2 + 3n) + m(n - 3n^2) + 2n^3 = 0$$

If $n = 0$, m can clearly be anything looking at original equation. Otherwise,

$$m^2n + 3m^2 + m - 3mn + 2n^2 = 0$$

$$\Leftrightarrow 2n^2 + (m^2 - 3m)n + (3m^2 + m) = 0$$

We look at the discriminant, which would be,

$$(m^2 - 3m)^2 - 4(2)(3m^2 + m)$$

$$= m(m^3 - 6m^2 - 15m - 8)$$

$$= m(m + 1)^2(m - 8)$$

Since this must be a perfect square, either $m = -1$, which gives $n = -1$, or,

$$m(m - 8) = (m - 4)^2 - 16$$

Is a perfect square. For $|m-4| > 5$ this would be between $(m-4)^2$ and $(m-3)^2$. For $|m-4| < 4$ this would be negative. Therefore, we have $m = -1, 0, 8, 9$. Plugging this in, we get $(-1,-1), (0,0), (8,-10), (9,-6), (9,-21)$ as desired.

Clearly, if $n > 0$, then the quantity inside the radical is negative.

4;6

Problem 5 (Pixton)

Sketch: Take mod 7 to get $y \geq 7$ fails. Then check $y = 4$ only solution.

Problem 6 (19USEMO4)

We plug in $n = p(p-1)(p^{1434} - 2020)$. Note that, since $k^{n-k} \equiv \left(k + p(p-1)\right)^{n-(k+p(p-1))} \bmod p$,

$$\sum_{i=1}^{n} i^{n+1-i} \equiv (p^{1434} - 2020) \sum_{i=1}^{p(p-1)} i^{p(p-1)+1-i}$$

$$\equiv (-2020)\left(\sum_{i=1}^{p} i^0 + i^1 + i^2 + \cdots + i^{p-2}\right)$$

$$\equiv (-2020)\left(\sum_{i=1}^{p-1} i^0 + i^1 + i^2 + \cdots + i^{p-2}\right)$$

$$\equiv (-2020)\left(\sum_{i=1}^{p-1}\left(i^0 + \frac{i^{p-1} - 1}{i-1}\right)\right)$$

$$\equiv (-2020)\left(\sum_{i=1}^{p-1} i^0\right)$$

$$\equiv 2020$$

9;6.5

Problem 8 (07BRA2)

Claim: A number $0 \leq x < 2^k$ is an odd quadratic residue mod 2^k if and only if $x \equiv 1 \bmod 8$ for all positive $k \geq 3$.

We prove a stronger conclusion that, for any k, $\left\{1^2, 3^2, \dots, \left(2^{k-2}-1\right)^2\right\} \equiv \{1, 9, 17, \dots, 2^k - 7\} \bmod 2^k$ by induction.

Base case: $k = 3$.

$$\{1^2\} \equiv \{1\} \bmod 2^5$$

Inductive step: $k = m \Longrightarrow k = m + 1$

First, note that, for any odd i,

$$(2^{m-1} - i)^2 \equiv i^2 - 2^m i + 2^{2(m-1)} \equiv i^2 - 2^m i \not\equiv i^2 \bmod 2^{m+1}$$

By the inductive hypothesis,

$$\{1^2, 3^2, \dots, (2^{m-2} - 1)^2\} \equiv \{1, 9, 17, \dots, 2^m - 7\} \bmod 2^m$$

Using our fact,

$$\{(2^{m-2} + 1)^2, (2^{m-2} + 3)^2, \dots, (2^{m-1} - 1)^2\} \equiv \{1, 9, 17, \dots, 2^m - 7\} \bmod 2^m$$

However, $i^2 \not\equiv (2^{m-1} - i)^2$ and obviously i^2 won't have the same residue as any other squares. Therefore, by Pigeonhole, every residue that is $1 \bmod 8$ will be covered. [citation needed?]

$$\{1^2, 3^2, \dots, (2^{m-1} - 1)^2\} \equiv \{1, 9, 17, \dots, 2^{m+1} - 7\} \bmod 2^{m+1}$$

Conclusion: By induction we're done.

Now casework on $v_2(c)$. We must have $v_2(x) = \frac{v_2(c)}{2}$ then plug in $X = \frac{x}{v_2(x)}$ to get $X \equiv 1 \bmod 8$. Since 2007 is small this should be fast.

Problem 10 (Z65C2983)

We claim that all $n \equiv 1, 3 \bmod 8$ work.

Claim: $3^m \equiv n \bmod 2^k$ for any $k \geq 3$ has a solution for m if and only if $n \equiv 1, 3 \bmod 8$.

The only if is trivial since $3^m \equiv 1, 3 \bmod 8$. For the if, we may prove that $v_{2^k}(3) = 2^{k-2}$ and use Pigeonhole.

Sketch of proof: Same induction process as above.

Base case for $n = 3$ is easy.

Use Euler's totient to get $v_{2^k}(3) \mid 2^{k-1}$. However, use inductive hypothesis to get $3^m \bmod 2^k$ does not repeat for $m \leq 2^{k-2}$ since mod 2^{k-1}. Additionally, if $3^{2^{k-2}} \equiv 1 \bmod 2^k$, use LTE to get this is impossible.

The claim immediately gives answer since if $n \not\equiv 1, 3 \bmod 8$ we cannot have $3^m - n \equiv 0 \bmod 2^k$ for any $k \geq 3$. If $n \equiv 1, 3 \bmod 8$ we can always get it.

Problem 12 (98SLN5)

We claim the answer is all $n = 2^k$ for some non-negative integer k.

We first show there is a construction for such n. Equivalently, we show that there exists a positive m such that,

$$m^2 \equiv -9 \bmod 2^{2^k} - 1$$

The case for $k = 0$ is trivial. For other k, note that,

$$\gcd\left(2^{2^{k-1}} - 1, 2^{2^{k-1}} + 1\right) = \gcd\left(2, 2^{2^{k-1}} - 1\right) = 1$$

Therefore,

$$\Leftrightarrow m^2 \equiv -9 \left(\bmod 2^{2^{k-1}} - 1\right);\ m^2 \equiv -9 \left(\bmod 2^{2^{k-1}} + 1\right)$$

Inducting, we show that there exists a m satisfying,

$$m^2 \equiv -9 \left(\bmod 2^{2^{k-1}} + 1\right) \Leftarrow m \equiv -3 \cdot 2^{2^{k-2}} \left(\bmod 2^{2^{k-1}} + 1\right);$$

$$m^2 \equiv -9 \left(\bmod 2^{2^{k-2}} + 1\right) \Leftarrow m \equiv -3 \cdot 2^{2^{k-3}} \left(\bmod 2^{2^{k-2}} + 1\right);$$

$$\dots$$

$$m^2 \equiv -9 \left(\bmod 2^{2^0} + 1\right) \Leftarrow m = -3 \left(\bmod 2^1 + 1\right)$$

By the Chinese remainder theorem, since all the mod bases are relatively prime with each other, we may select such a m.

We now show all other n fail.

Claim: All odd $n > 1$ fail.

We want,

$$2^n - 1 \mid m^2 + 9$$

Note that $2^n - 1 \not\equiv 0 \bmod 3$ but $2^n - 1 \equiv 3 \bmod 4$ so there's an odd prime other than 3 dividing $2^n - 1$ and hence dividing $m^2 + 3^2$. By Fermat's Christmas Theorem impossible.

With the claim proven, we may just notice, if $n \neq 2^k$ for some k, we must have, whenever n is even.

$$m^2 \equiv -9 \left(\bmod 2^{\frac{n}{2}} - 1\right); m^2 \equiv -9 \left(\bmod 2^{\frac{n}{2}} + 1\right)$$

Do this repeatedly to the left congruence until the exponent is an odd number greater than one.

Problem 14 (05SLN4)

We claim that the solution is all prime n.

Proof they work:

For greater n, we have this is true if and only if, for all primes p, where k is $v_p(n!)$,

$$a^n \equiv -1 \bmod p^k$$

$$\Longrightarrow a^{2n} \equiv 1 \bmod p^k$$

If $p^k = 2$, then $a \equiv -1 \bmod p^k$ is the only solution. Note that $ord_{p^k}(a) \mid 2n, \phi(p^k)$. For all other p^k, since n a prime greater than or equal to p^k,

$$\gcd\left(2n, \phi(p^k)\right) = \gcd(2n, p^k - p^{k-1}) = \gcd(2, p^k - p^{k-1}) = 2$$

Therefore, $ord_{p^k}(a) \mid 2$. We have $a \equiv \pm 1 \bmod p^k$ but positive clearly does not work. Therefore, we always have,

$$a \equiv -1 \bmod p^k$$

By the Chinese Remainder Theorem, there is exactly one solution, which is $a \equiv -1 \bmod n!$ so $a = n! - 1$.

Proof that all others fail:

For even $n > 2$, note that there are no solutions. We write $n = 2k$.

$$n! \mid a^n + 1 \Longrightarrow 3 \mid \left(a^k\right)^2 + 1^2$$

By Fermat's Christmas Theorem this is impossible.

For odd n, note that $a = n! - 1$ is still a solution. We claim that $a = (n-1)! - 1$ is also a solution. For any $p^k \parallel n$,

$$\left((n-1)! - 1\right)^n + 1 \equiv 0 \bmod p^k$$

$$\Leftrightarrow v_p\left(\left((n-1)! - 1\right)^n + 1\right) \geq k$$

$$\Leftrightarrow v_p\left(\left((n-1)! - 1\right)^n + 1^n\right) \geq k$$

Note that $(n-1)! - 1 + 1 = (n-1)! \mid p$ since n is composite. Therefore, since n is odd, we may apply lifting the exponent to get,

$$\Leftrightarrow v_p\left((n-1)!\right) + v_p(n) \geq k$$

$$\Leftrightarrow v_p(n!) \geq k$$

By the definition of $p^k \parallel n!$, this is true.

Problem 15 (07SLN5)

We claim that $f(x) = x$ is the answer, which clearly works.

Note that $f(2)$ shares prime divisors with $f(1) + f(1) = 2$, so $f(2) = 2^k$ for some k. Inducting, f of all powers of two outputs a power of two.

Since $f(4)$ shares divisors with $f(1) + f(3) = f(3) + 1$, $f(3) + 1$ is a power of two, so, for some x, y, z,

$$f(1) = 1; f(2) = 2^x;\ f(3) = 2^y - 1; f(4) = 2^z$$

Therefore, $2^x + 1$ shares prime divisors with $2^y - 1$.

For $y = 2x$, we'd get $2^x + 1 \mid 2^y - 1$, $2^x - 1 \mid 2^y - 1$. The only way this is possible is if $2^x - 1 = 1$ and $x = 1, y = 2$.

By Zsigmondy, in every other case, there exists a prime $p \mid 2^{2x} - 1$ but p does not divide $2^y - 1$ or vice versa. In the exception case where $x = 3$, we have $y = 2$ but then $f(2) = 8$ and $f(3) = 3$ which leads contradiction looking at $f(4)$.

Therefore,

$$f(1) = 1; f(2) = 2; f(3) = 3$$

Lemma: If $2^x - 1$ and $2^y - 1$ shares prime divisors, $x = y$. For $2^x + 1$ and $2^y + 1$, $x = y$ or $(x, y) = (1, 3)$ or $(3, 1)$.

Proof: This is Zsigmondy. Exceptions such as $x = 6$ in first case trivially resolve.

Since $f(5)$ shares prime divisors with both $f(2) + f(3) = 2^2 + 1$ and $f(1) + f(4) = 2^k + 1$ for some k where $f(4) = 2^k$. Once again, by lemma, $k = 2$.

We claim that $f(x) = x$ for $2^e < n \leq 2^{e+1}$ using induction on e, which finishes.

Base case: Done.

Inductive step:

We have $f(x) = x$ for all $x \leq 2^e$.

We first prove that $f(2^{e+1} - 1) = 2^{e+1} - 1$. We already have that $f(2^{e+1} - 1) = 2^k - 1$ for some k. However, $f(2^{e+1} - 1)$ shares divisors with $f(2^e - 1) + f(2^e) = 2^{e+1} - 1$. Using the lemma, we are done.

We next prove that $f(2^{e+1}) = 2^{e+1}$. Similar as before, let $f(2^{e+1}) = 2^k$ and look at $f(2^{e+1} + 1)$ to get $k = e + 1$.

We now finally prove the rest, that $f(x) = x$ for $2^e + 1 \leq x \leq 2^{e+1} - 2$. We have,

$f(x) + f(2^{e+1} - x) = f(x) + 2^{e+1} - x$ shares prime divisors with 2^{e+1}, or 2. Therefore, for some k,

$$f(x) + 2^{e+1} - x = 2^k$$

Similarly, $f(x) + f(2^{e+1} - x - 1) = f(x) + 2^{e+1} - x - 1 = 2^k - 1$ shares prime divisors with $2^{e+1} - 1$. Therefore, $k = e + 1$, $f(x) = x$ as desired.

36;12.5

<u>Problem 19 (90IMO3)</u>

We claim that $n = 1, 3$ is the only solution, which works. Consider $n > 1$.

Clearly, even n fail since $2^n + 1$ is odd.

Claim: $3 \mid n$

Consider the smallest prime p dividing n.

$$2^n \equiv -1 \bmod p$$

$$\Longrightarrow 2^{2n} \equiv 1 \bmod p$$

Therefore, $\operatorname{ord}_p(2) = \gcd(p - 1, 2n)$. Since p is the smallest prime, $\gcd(p - 1, n) = 1$. On the other hand, since p is odd, $p - 1$ is even. Therefore,

$$\operatorname{ord}_p(2) = 2$$

We hence have,

$$2^2 \equiv 1 \bmod p$$

Clearly, $p = 3 \mid n$.

Claim: $v_3(n) = 1$

Since n is odd, we may use LTE to get,

$$v_3(n^2) \leq v_3(2^n + 1)$$

$$\Leftrightarrow 2v_3(n) \leq v_3(3) + v_3(n)$$

$$\Leftrightarrow v_3(n) \leq 1$$

Since $3 \mid n$ we are done.

Now consider $n > 3$. There must be a p' the second smallest prime dividing n since 2 cannot divide n and 9 cannot divide n. We similarly have,

$$2^{2n} \equiv 1 \bmod p'$$

$$\Longrightarrow \operatorname{ord}_{p'}(2) = \gcd(p - 1, 2n)$$

We must have $3 \mid p - 1$ for if otherwise the gcd is 2 and we have $p' = 3$ as before. In this case,

$$\gcd(p - 1, 2n) = 6$$

Therefore,

$$2^6 = 64 \equiv 1 \bmod p'$$

Clearly, since $p' \neq 3$, we have $p' = 7$. However, $2^n \equiv 1, 2, 4 \bmod 7$ so $7 \mid 2^n + 1$ is impossible.

Problem 20 (00IMO5)

I cannot find another identical solution on AoPS so idk if this works.

We claim that this is possible. We give a constructive proof using induction on the number of distinct prime divisors. In the induction, we prove that, for any integer x, we may find an integer n_x such that,

$$n_x \mid 2^{n_x} + 1$$

Additionally, we require that, also, for $x > 1$, $n_x \mid 2^{n_{x-1}} + 1$.

Base cases:

$x = 1$, we can have $n_1 = 9$.

$x = 2$, we can have $n_2 = 9 \cdot 19$, which works since we can check,

$$n_2 \mid 2^{n_1} + 1 \mid 2^{n_2} + 1$$

Inductive Step:

Given,

$$n_{x-1} \mid 2^{n_{x-1}} + 1$$

We claim that it is possible to find an odd prime $p \nmid n_{x-1}$ such that, also,

$$p \mid 2^{n_{x-1}} + 1$$

Using the inductive hypothesis, $n_{x-1} \mid 2^{n_{x-2}} + 1$, so by Zsigmondy, there must exist this new prime. We then have,

$$pn_{x-1} \mid 2^{n_{x-1}} + 1 \mid 2^{pn_{x-1}} + 1$$

We can have $n_x = pn_{x-1}$.

Conclusion: By induction we have $x = 2000$ works also.

44;15

Problem 21 (18TSTST8)

We claim that the only solutions are $b \neq 2^k - 1$ for some k.

Proving this works:

We construct infinite solutions, which are, for some primes $p_1, p_2, \ldots$, we may have,

$$n = p_1, p_1p_2, p_1p_2p_3, \ldots$$

We use induction to prove that we may find such a sequence of primes which satisfy for any m,

$$p_1^2 \dots p_m^2 \mid b^{p_1 p_2 \dots p_m} + 1$$

Also, we require that, $p_1^2 \dots p_{m-1}^2 p_m \mid b^{p_1 p_2 \dots p_{m-1}} + 1$.

Base case: $m = 1$

Let p_1 be any prime $p_1 \mid b + 1$. Using LTE, we have,

$$p_1^2 \mid b^{p_1} + 1$$

We consider $p_1 p_2 \dots p_{m-1} = 1$ for $m = 1$, we defined $p_1 \mid b + 1$ so the second condition is met.

Inductive step:

Given,

$$p_1^2 p_2^2 \dots p_{m-2}^2 p_{m-1} \mid b^{p_1 p_2 \dots p_{m-2}} + 1$$

By Zsigmondy, there exists a new prime p_m dividing $b^{p_1 p_2 \dots p_{m-1}} + 1$. Therefore,

$$p_1^2 p_2^2 \dots p_{m-2}^2 p_{m-1}^2 p_m \mid b^{p_1 p_2 \dots p_{m-1}} + 1$$

By lifting the exponent,

$$p_1^2 p_2^2 \dots p_{m-2}^2 p_{m-1}^2 p_m^2 \mid b^{p_1 p_2 \dots p_{m-1} p_m} + 1$$

Proving $b = 2^k - 1$ *fails:*

Consider $n > 1$,

Let p be the smallest prime dividing $b^n + 1$,

$$b^n \equiv -1 \bmod p$$

$$\Leftrightarrow b^{2n} \equiv 1 \bmod p$$

Using orders,

$$b^2 \equiv 1 \bmod p$$

$$p \mid (b - 1)(b + 1)$$

However, $b + 1 = 2^k$, so $p \mid b + 1$ is impossible. Therefore,

$$p \mid b - 1$$

We hence have,

$$b^n + 1 \equiv 2 \bmod p$$

But $p \neq 2$ so this impossible.

Problem 23 (19IMO4)

The solutions are $(n, k) = (1,1), (2,3)$.

Plugging in $n = 1,2,3,4$, we respectively get $k = 1, 3, \text{impossible}, \text{impossible}$.

Claim: $k > \frac{n(n-1)}{2}$.

Taking v_2,

$$k > v_2(k!) = v_2(2^n - 1) + v_2(2^n - 2) + \cdots + v_2(2^n - 2^{n-1})$$

$$\Longrightarrow k > \frac{n(n-1)}{2}$$

Claim: $k < 2n$ *for* $n > 4$

Taking v_3,

$$v_3(k!) = v_3(2^n - 1) + v_3(2^n - 2) + \cdots + v_3(2^n - 2^{n-1})$$

$$\Longleftrightarrow v_3(k!) = v_3(2^n - 1) + v_3(2^{n-1} - 1) + \cdots + v_3(2^1 - 1)$$

Note that $2^n \equiv 1 \bmod 3$ if and only if n is even.

$$\Longleftrightarrow v_3(k!) = v_3\left(4^{\left\lfloor\frac{n}{2}\right\rfloor} - 1\right) + v_3\left(4^{\left\lfloor\frac{n}{2}\right\rfloor - 1} - 1\right) + \cdots + v_3(4^1 - 1)$$

Using lifting the exponent,

$$\Longleftrightarrow v_3(k!) = \left\lfloor\frac{n}{2}\right\rfloor + v_3\left(\left\lfloor\frac{n}{2}\right\rfloor\right) + v_3\left(\left\lfloor\frac{n}{2}\right\rfloor - 1\right) + \cdots + v_3(1)$$

$$\Longleftrightarrow v_3(k!) = \left\lfloor\frac{n}{2}\right\rfloor + v_3\left(\left\lfloor\frac{n}{2}\right\rfloor!\right)$$

Note that if $k \geq 2n$, this is impossible as, on this range, [citation needed?]

$$v_3\left(\frac{k!}{\left\lfloor\frac{n}{2}\right\rfloor!}\right) \geq v_3\left(\frac{2n!}{\left\lfloor\frac{n}{2}\right\rfloor!}\right) > \left\lfloor\frac{n}{2}\right\rfloor$$

Therefore, $k < 2n$ as desired.

We then have,

$$2n > k > \frac{n(n-1)}{2}$$

$$\Longrightarrow 2n > \frac{n(n-1)}{2}$$

$$\Longleftrightarrow 4 > (n-1)$$

$$\Longleftrightarrow n < 5$$

We are already done with this range.

56;17.5

<u>Problem 26 (21USEMO2)</u>

We claim that $n = 1, 2, 4, 6, 8, 16, 32$ are the only solutions. We will prove that these work and that these are the only solutions as follows.

Claim: If $n \neq 1$ then n is even.

If not, then n is odd and $2^n - 1$ is a square, which is impossible for $n > 1$ since $2^n - 1 \equiv 3 \bmod 4$.

We write, $n = 2^a b$ for some positive a and some odd number b.

We have that the following quantity has $2^a b$ factors.

$$2^n - 1 = (2^b - 1)(2^b + 1)(2^{2b} + 1) \dots (2^{2^{a-1}b} + 1)$$

Claim: $b \in \{1, 3\}$

If not, then $b \geq 5$. We have $2^b - 1 \equiv 3 \bmod 4$ cannot be a square. We claim that $2^{2^k b} + 1$ cannot be a square either. FTSOC, if $2^{2^k b} + 1 = a^2$,

$$(a - 1)(a + 1) = 2^{2^k b}$$

We have $a - 1, a + 1$ must both be powers of two that multiply to at least 32, which is impossible.

We hence have $(2^b - 1), (2^b + 1), (2^{2b} + 1), \dots, (2^{2^{a-1}b} + 1)$ are not squares so each have an even number of divisors. *Insert proof they're relatively prime each other* so the number of factors the product has is a multiple of 2^{a+1}. (!)

Case 1: $b = 1$

We have,

$$2^n - 1 = (1)(2^1 + 1) \dots (2^{2^{a-1}} + 1)$$

Has 2^a factors. Since relatively prime, this is possible if and only if $(2^1 + 1), \dots, (2^{2^{a-1}} + 1)$ are prime. These are the Fermat primes. Using the famous fact that this is true up to $a - 1 = 4$, we have $n = 1, 2, 4, \dots, 32$.

Case 2: $b = 3$

$$2^n - 1 = (7)(9)(2^{2\cdot 3} + 1) \dots (2^{2^{a-1}\cdot 3} + 1)$$

Similarly, using relatively prime and the fact that 7 has two divisors with 9 having 3, we must have this is possible if and only if $(2^{2\cdot 3} + 1), \dots, (2^{2^{a-1}\cdot 3} + 1)$ are prime. However, $2^6 + 1 = 65$ is not prime so $a > 1$ does not work. The only solution is $a = 1$.

61;19